FR ES EN

iBlueDrive Technology user's Manual

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Manual & specifications for iBlueDrive devices

CONTENT

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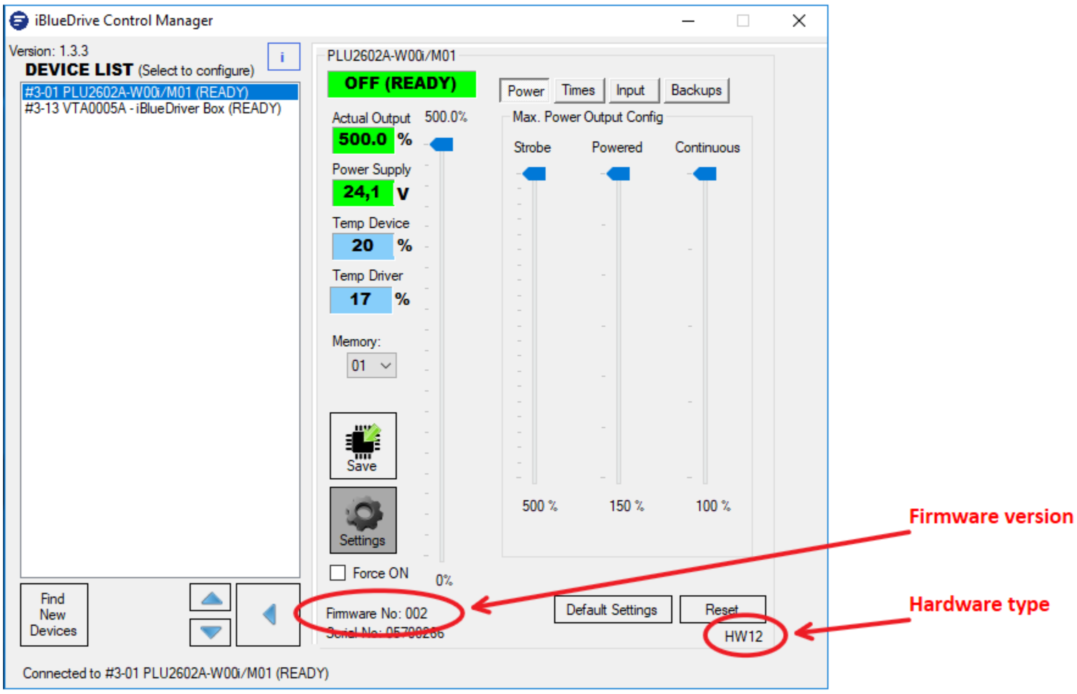
1. Scope

The purpose of this manual is to describe all general specifications and explain the operating functions of **©iBlueDrive** lighting devices. **©iBlueDrive** is a technology property of DCM Sistemes, which allows an easy and powerful use of LED lighting systems in machine vision applications.

This manual is only applicable to the following iBlueDrive version hardware and firmware:

Hardware_ID	Firmware_Version	Description
12	006	New version of iBlueDrive lightings

Hardware_ID and ***Firmware_Version*** can be found in the panel of control at the **iBlueDrive Control Manager** application.



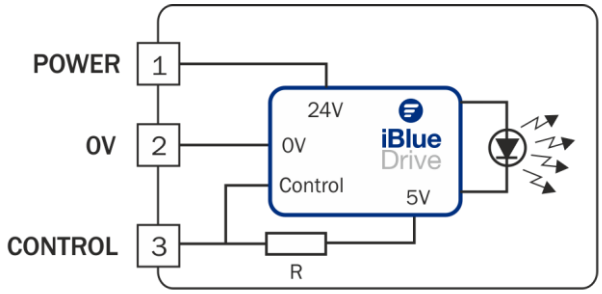
2. Specifications highlights

- Power Supply:
    - Operating Voltage: **24VDC ± 8%**
    - Standby consumption: **<720mW**, 550mW typical consumption.
    - Max Consumption: Consult product label.
- Wiring: 3 contacts female M9 circular connector (VCC series cable required)
- Operating temperature = **0-40 ºC**
- Storage temperature = **0-60 ºC**
- RoHs, CE, WEEE regulations

3. Special ©iBlueDrive features

- Automatic strobe, powered, continuous working mode.
- Easy wiring. Only three terminals.
- Direct connection to camera or trigger signal.
- Automatic NPN or PNP detection.
- No configuration needed. Preset to shutter speeds of up to 47 pulses per second (pps)
- Shutter speed up to 5000 Hz (requires setting with control software).
- Easy dimming control in all modes.
- Configuration offline and online. Set power output, pulse width, delays, etc. with intuitive software (requires additional hardware).
- Dynamic blinking LED indicator. Informs about status and working temperature.
- All information at your fingertips. Input voltage, driver and device working temperature, triggering mode, serial number among others with our software communication (requires additional hardware).

4. Simplified circuit

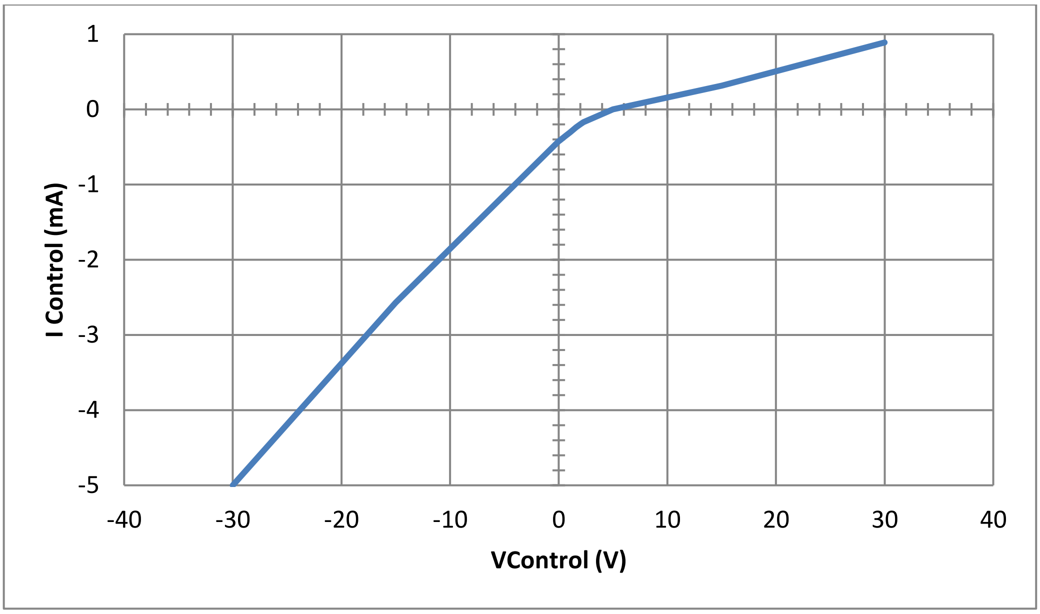


5. Electrical specifications

DC Characteristics					
Standard Operating Conditions (unless otherwise stated). Operating temperature 0°C < TA < +40°C					
Symbol	**Characteristic**				
	Min.	**Typ.**	**Max.**	**Units**	**Conditions**
Vnom	**Nominal supply voltage to work properly**				
	22.1	24.0	26.0	V	
Vmax	**Max. voltage allowed in Vpower regarding to 0V terminal**				
	-30.0	-	30.0	V	Permanent condition
VCdFull	**Voltage at control terminal to get 100% of continuous power in dimmer mode**				
	0.75	0.8		V	
VCdOff	**Voltage at control terminal to turn OFF the output in dimmer mode**				
		1.75	1.8	V	
Inom	**Nominal working current**				
		15	25	mA	Powered with 24VDC, light Off
VCz	**Voltage at control terminal with free terminal**				
	4.9	5	5.1	V	
VCd	**Voltage range at control terminal in dimmer mode**				
	0		3	V	
VCd_100%	**Voltage at control terminal in dimmer mode for 100% continuous power output**				
		0.75		V	
VCd_0%	**Voltage at control terminal in dimmer mode for 0% continuous power output**				
		1.75		V	
VCdExit	**Voltage at control terminal to get automatically out of dimmer mode**				
	3.5			V	
VControl	**Allowable voltage at control terminal**				
	-30		30	V	
VCoff	**Voltage at control terminal to turn OFF the light**				
	2.6		8	V	
VCon_Lw	**Voltage at control terminal to turn ON the light (NPN mode)**				
			1.6	V	
VCon_Hi	**Voltage at control terminal to turn ON the light (PNP mode)**				
	9			V	
IC_0V	**Current at control terminal when connecting to 0V**				
		-0.4		mA	Sink, device powered with 24VDC
IC_24V	**Current at control terminal when connecting to 24V**				

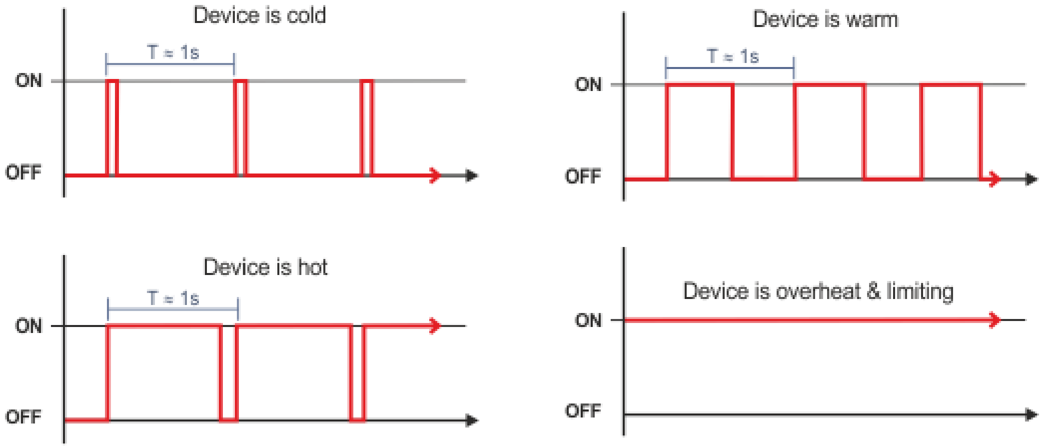
DC Characteristics					
Standard Operating Conditions (unless otherwise stated). Operating temperature 0°C < TA < +40°C					
Symbol	**Characteristic**				
	Min.	**Typ.**	**Max.**	**Units**	**Conditions**
		0.7		mA	Sink, device powered with 24VDC
TH_Op	**Operating temperature**				
	0		40	ºC	
H_Op	**Operating humidity**				
			80	%	Non condensing
TH_St	**Storage temperature**				
	0		60	ºC	
T_On	**Turn ON delay regarding to control signal**				
	33.6	50		µs	Typ. Is for factory settings
T_On_BT	**Turn ON delay regarding to control signal when Bulk Trigger is enabled**				
	48	60		µs	Typ. Is for factory settings
T_Tol	**Internal timer tolerance**				
		±2		%	
T_CSW	**Minimum input Control Signal Width**				
	0.5	28.2	219	µs	Typ. Is for factory settings
T_Off	**Turn OFF delay regarding to control signal**				
	22	38		µs	Typ. Is for factory settings
TS_W	**Allowable strobe pulse width**				
	10		20,000	µs	No LED limits. The maximum pulse width is frequently reduced by LEDs' capability.
T_Slope	**Slope time for switching**				
			3	µs	
T_PowerON	**Time to get ready to work when turning ON the power**				
		980	1050	ms	

Current vs. Voltage applied in the control terminal



6. Dynamic blinking LED indicator

iBlueDrive lighting systems have a blue LED indicator that informs the user about the driver status and the thermal temperature of the device.



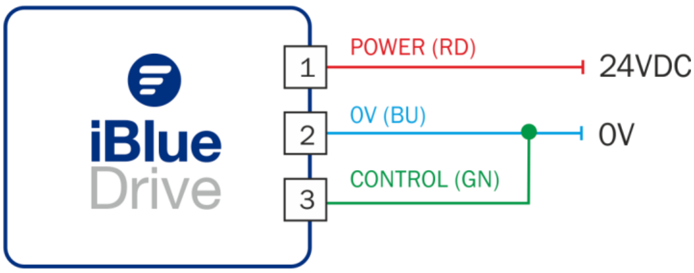
When the driver is running correctly, the LED blinks in intervals of around 1s. The time that LED status indicator is ON depends on internal temperature. When the light gets hotter, the time ON of the LED indicator is longer. When the status LED indicator is always ON, the device is overheated and it is limiting the output light to prevent burning.

7. Continuous mode operation

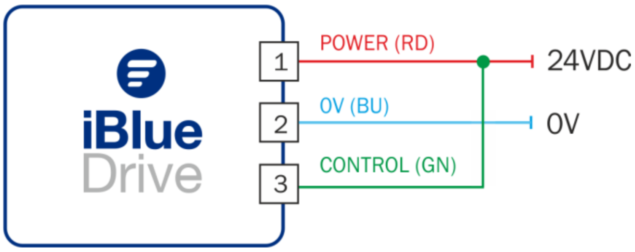
Continuous mode operation is the easiest mode of use, but it is also the less recommended because you cannot take advantage of the ©**iBlueDrive** technology.

Join the control terminal to 0V or 24V before Power ON to turn On the lighting in continuous mode.

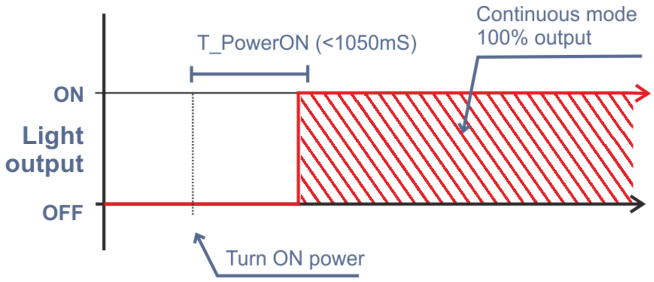
Wiring the control terminal to 0V:



Wiring the control terminal to 24V:



Timing chart at power ON:



CAUTION!
Continuous mode is the most power consuming and it is also the one that provides less illumination.
Some devices, specially the new developed iBlueDrive lamps with high density power and no heat sinks, must be fixed to a metal part of the machine or add a heat sink accessory, if available, to avoid overheat.
Check the blinking LED status indicator to check the lighting temperature or connect the lighting system to ©iBlueDrive software in order to check it.

8. ON/OFF or strobe mode operation

By activating or deactivating the signal at control terminal you can turn ON and OFF the ©**iBlueDrive** lighting device. That reduces the average power, cooling the light and allowing working in a special continuous mode named ON/OFF mode 'powered' mode, which provides up to 200% of the continuous mode power.

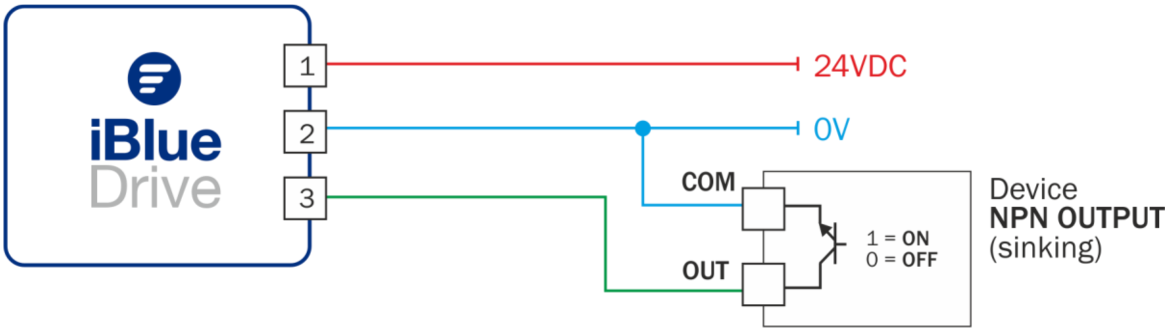
ON/OFF mode is used when very long exposure times are used in static piece inspection, such as palletizing.

When the switching ON/OFF is very fast, the ©**iBlueDrive technology** switches to Strobe mode automatically. In this mode, the light turns ON so much faster with a pre-programmed time (Pulse Width), and requires the use of a trigger signal to synchronize the lighting output with the camera's integration time. Trigger signal is frequently obtained from the camera or from a piece detector.

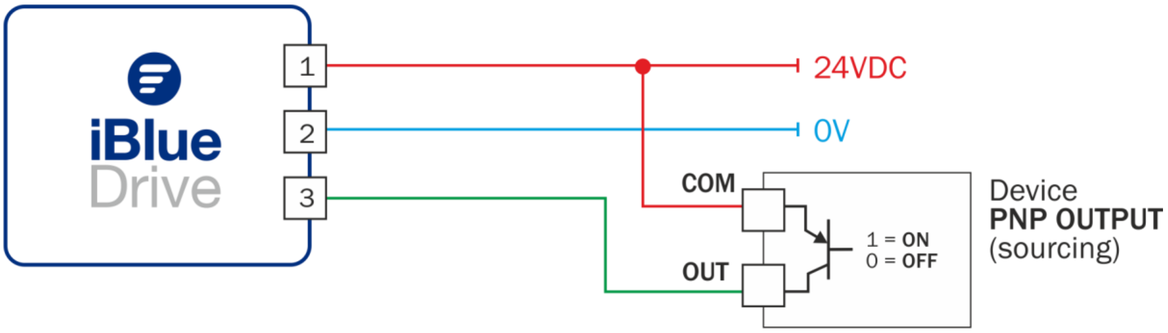
Working in strobe mode reduces drastically the average power and provides up to 800% or even more of the continuous mode power.

NPN, PNP or TTL signal can be used at control terminal without setting up. Autodetection input mode will set up the input type automatically.

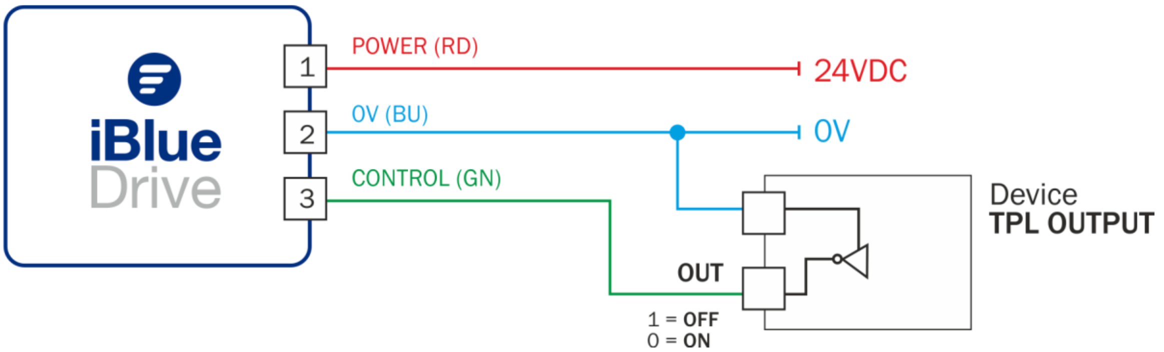
Wiring to a NPN output:



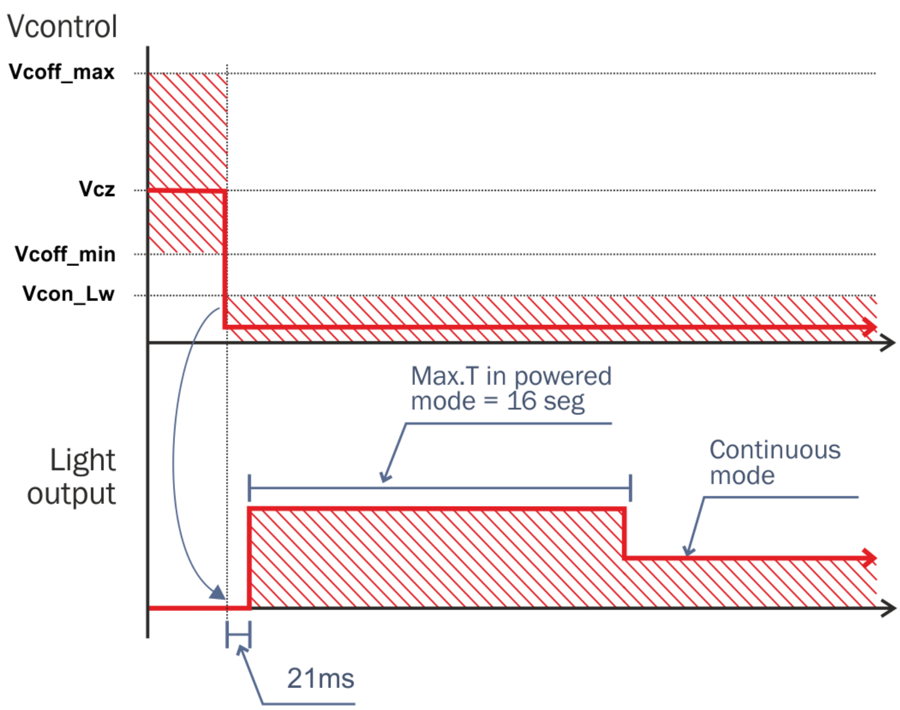
Wiring to a PNP output:



Wiring to a TTL output:



Timing chart for ON/OFF mode (NPN or TTL):



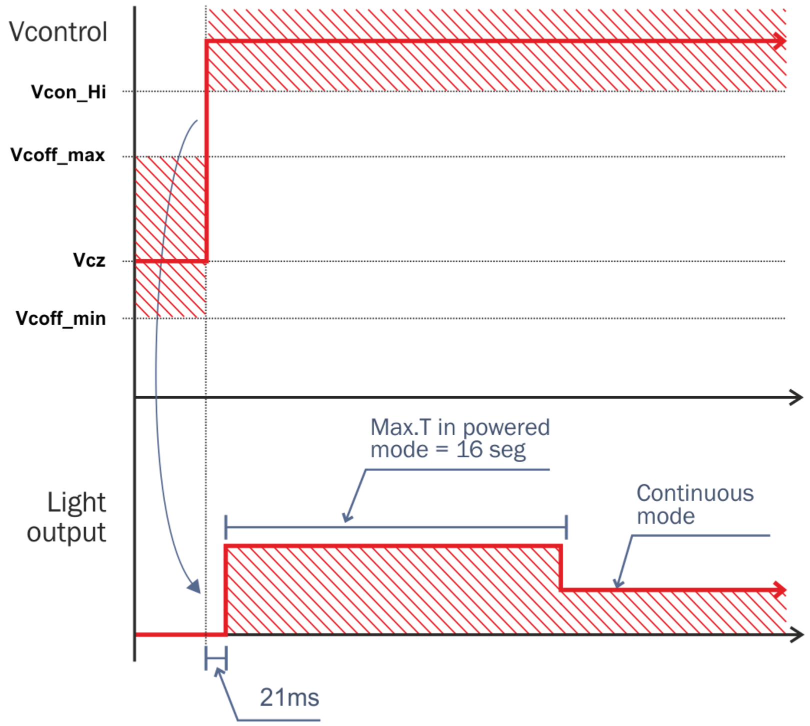
If the control terminal is connected to an NPN or TTL output, each time you force a low level in the control terminal, the light will turn ON with a 21ms delay time (*default settings*). NPN outputs no need a pull up resistor because **iBlueDrive** incorporates an internal voltage (Vcz) and a pull up resistor to work directly but pull up resistor should be recommended in noisy environments.

With control signals larger than 21ms, the light will turn ON in powered mode (default setting).

If the signal is kept low for more than 16 seconds, the lamp will reduce the light output to the normal continuous mode settings.

Each time the light is turned ON, the cycle will be repeated.

Timing chart for ON/OFF mode (PNP):



If the control terminal is connected to a PNP output, each time you force a Hi level in the control terminal, the light will turn ON with a 21ms delay time (*default settings*). A pull-down resistor is not needed because **iBlueDrive** incorporates an internal voltage (Vcz) and a pull up resistor to work directly.

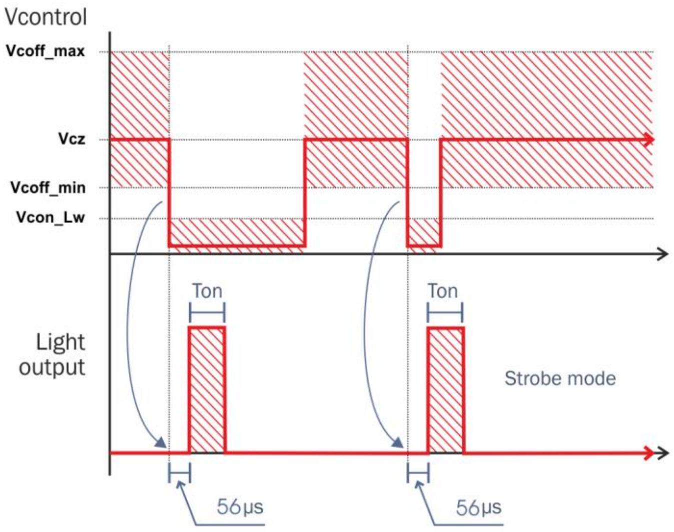
If a pull-down resistor is added, you must set the input type to PNP with the iBlueDrive Control Manager software. Otherwise the lighting could turn on in continuous mode.

With control signals larger than 21ms, the light will turn ON in powered mode (default setting).

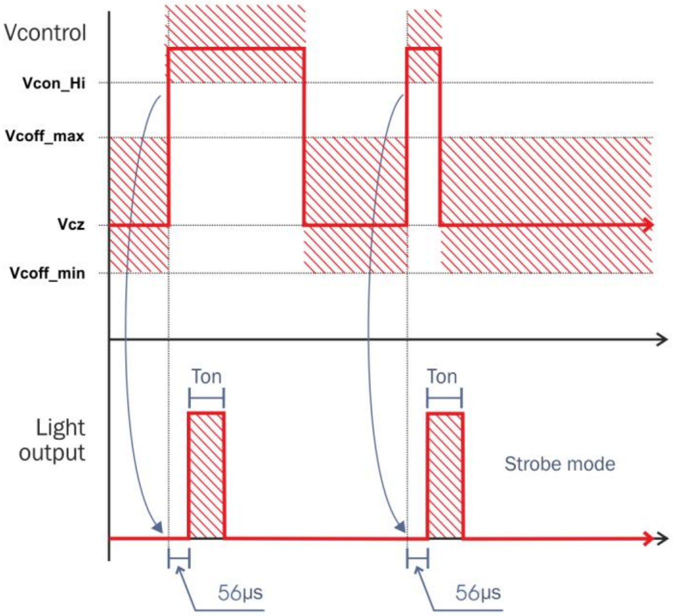
If the signal is kept active for more than 16 seconds, the lamp will reduce the light output to the normal continuous mode settings.

Each time the light is turned ON, the cycle will be repeated.

Timing chart for strobe mode (NPN or TTL):

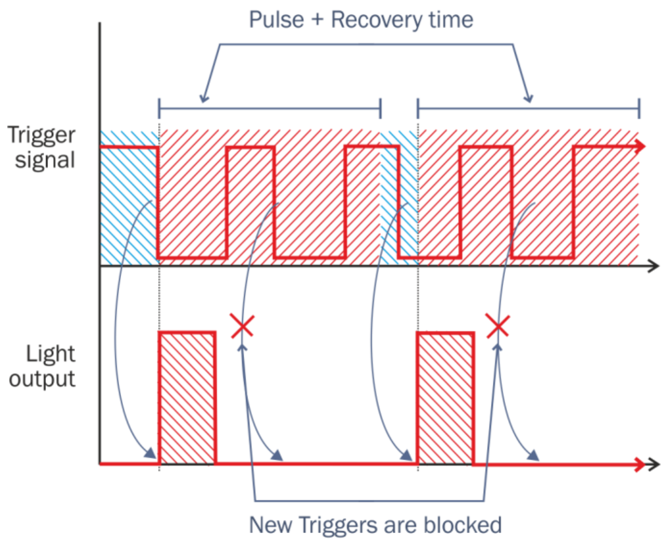


Timing chart for strobe mode (PNP):



When connecting the control terminal to a trigger output of a camera, PLC or a piece detector such as a photo-detector, the **iBlueDrive** lamp will be able to work in strobe mode automatically. With short activating signals, the light only outputs a strobe pulse of fixed width (Ton). Factory settings for the pulse width will vary from 2ms to 4.2ms depending on the lighting model. Consult the pulse width in the quick-guide attached to the lighting system.

When a strobe trigger is detected, no new triggering is allowed until the current pulse (Ton) and the recovery time have finished.



Recovery time is the period of time that LEDs have to be kept OFF after a strobe pulse, in order to allow a new strobe pulse without blowing out. It depends on the pulse width of the lamp's setting and the duty cycle (D) which is an internal parameter of the lighting device.

Therefore, the maximum Pulses Per Second (PPS) depend directly on the pulse width. With longer pulse widths (Ton), a lower maximum PPS is obtained and, on the contrary, with shorter pulse widths (Ton), is obtained higher maximum PPS. The following chart will show you an example of pulse width and max PPS available for a PLA1013A:

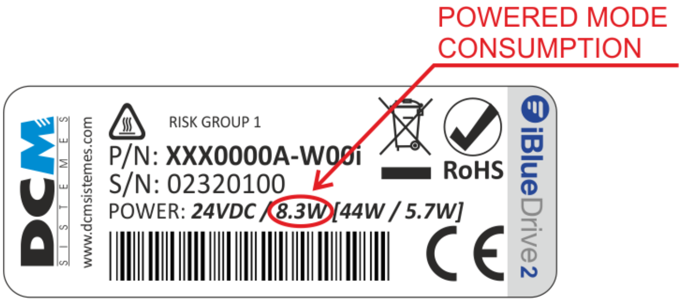
Pulse Width (Ton)	Max. Frequency available (PPS)	Note
20ms	10Hz	Lowest PPS for the max. Ton
10ms	20Hz	
4.2ms	47Hz	Default setting
1ms	194Hz	
200µs	1000Hz	
10µs	6000Hz	Highest PPS

iBlueDrive software will inform you about the recovery time for PPS allowed by the current settings. You can also consult these parameters for the factory settings in the quick-guide attached whit the ©**iBlueDrive** lighting system.

In order to avoid new triggers for a longer time, you can use the trigger OFF setting.

9. Power supply selection

Power supply must be selected depending on the total power consumption of all devices connected to it. Although the **iBlueDrive technology** can output and thereby spend a large quantity of power in strobe mode, the average power that has to be considered is the 'powered mode' consumption, because the average power in strobe mode should be lower due to the fact that the duty cycle is lower.

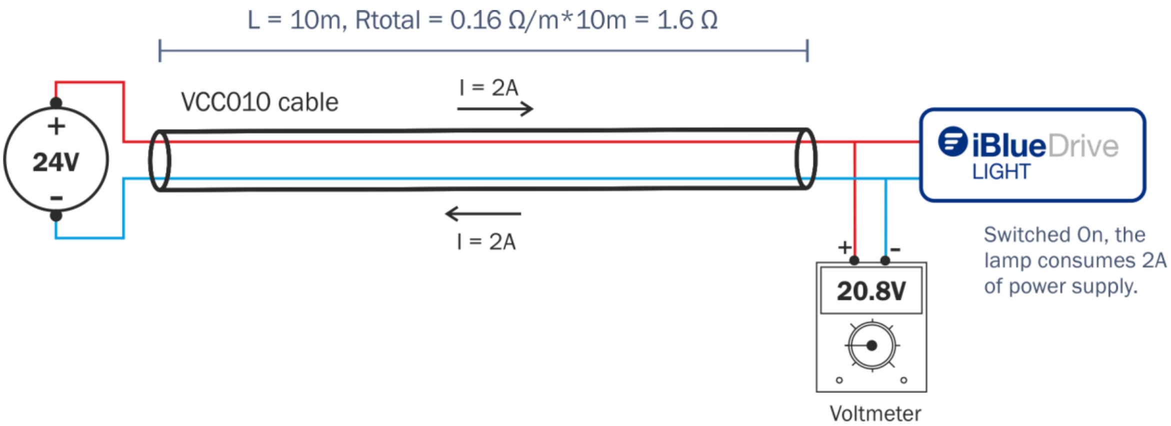


Powered mode consumption can be found in the product label of the lighting system.

10. Wiring

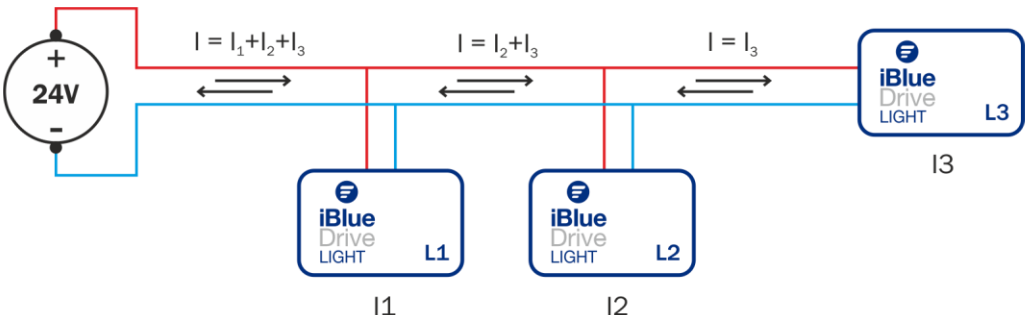
As **iBlueDrive** is an active driver, it is less sensible than other type of drivers to the voltage drops. However, it is a good working practice to use wires as shorter as possible. **iBlueDrive** can work with up to 20 meters cables, but cable impedance must be taken into account to not exceed the electrical limits.

iBlueDrive devices use the 'VCC Series' cable of DCM Sistemes, which has an impedance of 80mΩ/m per conductor or 160mΩ/m per cable (conductive meters through which current flows are usually twice the cable length).

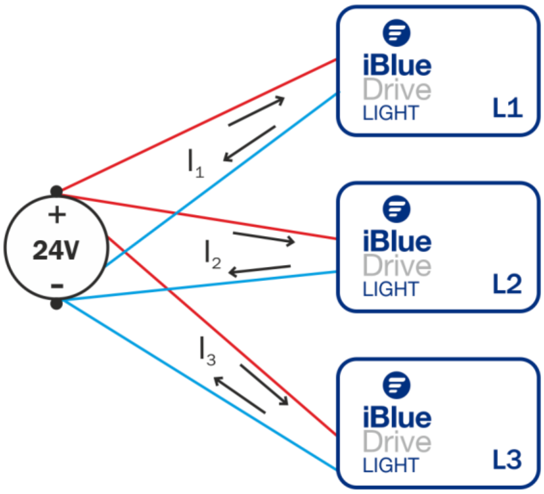


For example, with a lighting device rated at 48W, the current from power supply will be 48W/24V=2Amps. If we use a power cable ref. VCC010 that has 10m and 0.16Ohms/m, then we have a total impedance of 1.6Ohm (0.8Ohms per round) so the actual voltage in the light will be 20.8V, which is under the minimum permissible voltage.

Wiring topology is also a crucial factor when wiring, due to the fact that a wrong selection will suppose that lamps are incorrectly powered. Use a star topology, if possible, or a low impedance cable if it is required.



When wiring in bus topology, currents of the different lamps flow by the same cable, so that the voltage dropping in each lamp must be taken into account when all the lighting systems are ON.



With a star topology, cable currents are lower than in bus topology, but it frequently requires a greater total length of cable.

11. Advanced use (PC Communication)

Factory settings of ©**iBlueDrive** lighting devices fit the most of the industrial applications. However, it is sometimes required some extra settings and information that, with ©**iBlueDrive** technology, can be made ***'Off Line'***, when the application requires some setting changes in the light during the development time but not after, or ***'ON Line'***, when the application requires to change any settings of the light or to monitor the working, whilst inspection is running.

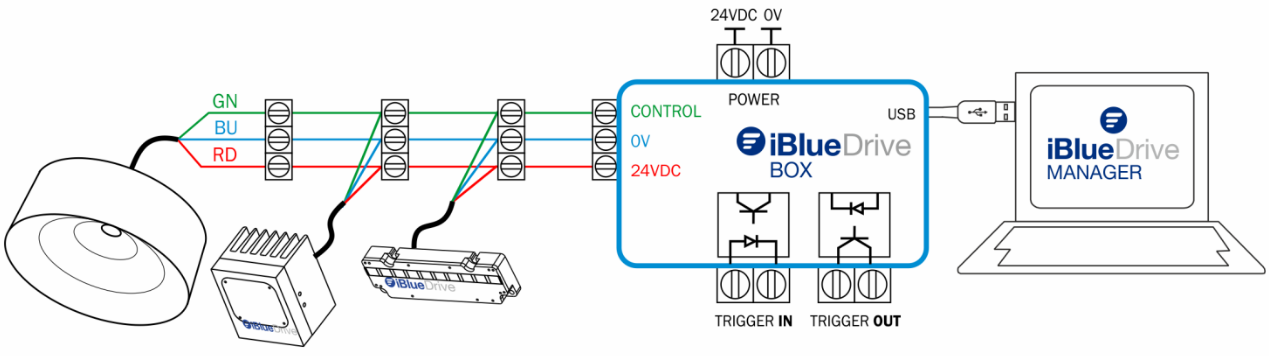
In order to connect the iBlueDrive lamp to the PC you can use:

USB cable **VTA0007A** for 'Off Line' use

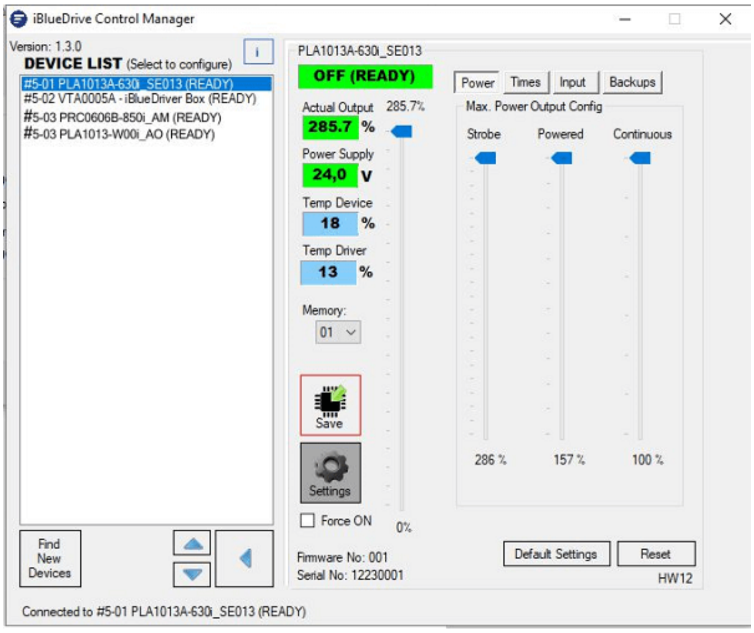
or

iBlueDrive Box **VTA0005A** and **VTA0006A** to work also 'On Line' use

It works as an interface between the USB port of the computer and the control terminal of the iBlueDrive lamp, mixing bidirectional communications with the trigger signals.



The free software iBlueDrive Control Manager (iBlueDrive_CM) is a very easy way to change settings and check the status and information of each connected iBlueDrive lamp.

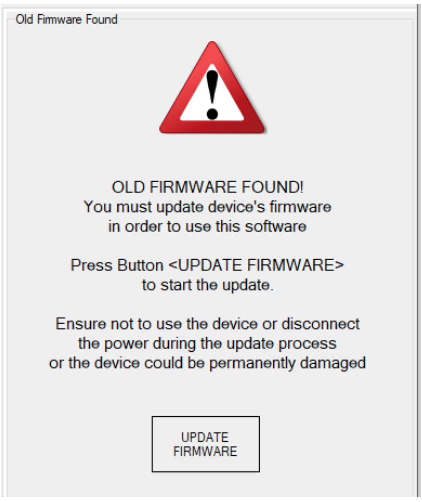


Refer to the iBlueDrive_CM manual for extended information.

12. Firmware updates

DCM Sistemes works continuously in the development of new functionalities in the ©i**BlueDrive** lighting systems that only requires a firmware update to be enjoyed.

Updating process of a firmware's lamp is very easy. Connect your iBlueDrive lighting to the last version of iBlueDrive_CM software through an iBlueDrive Box or iBlueDrive USB cable and follow the steps in the software.



If your lamp's firmware is out-of-date, the software will show you a warning advice. You can update the firmware by clicking the 'UPDATE FIRMWARE' button.

WARNING!
During the updating process, you must NOT trigger or disconnect the lighting system. Otherwise the lighting system could be seriously damaged.

13. Conclusion

Thanks for using **iBlueDrive** technology products of DCM Sistemes. Any doubt, clarification or suggestion you kindly have for improving our products, please address it to info@dcmsistemes.com and our technical engineers will effectively handle your requests.